

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech., Winter 2017 - 18 Examination

Semester: 5**Subject Code: 03105302****Subject Name: Design and Analysis of Algorithm****Date: 23/12/2017****Time: 10:30am to 1:00pm****Total Marks: 60****Instructions:**

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1 Objective Type Questions.**(15)****A Fill in the Blanks.**

1. Running merge sort on an array of size 'n' which is already sorted is _____.
2. Time complexity of bubble sort in best case is _____.
3. A _____ tree is a subset of Graph G, which has all the vertices covered with minimum possible number of edges.
4. _____ is a complexity class that represents the set of all decision problems for which the instances where the answer is "yes" have proofs that can be verified in polynomial time.
5. Worst case time complexity for quick sort is _____

B Choose the Correct Option.

1. Two main measures for the efficiency of an algorithm are
 - a. Processor and memory
 - b. Complexity and capacity
 - c. Time and space
 - d. Data and space
2. Let 'A' be a problem that belongs to the class NP. Then which one of the following is TRUE?
 - a. There is no polynomial time algorithm for 'A'
 - b. If 'A' is NP-hard problem, then it is NP-Complete
 - c. If 'A' can be solved deterministically in polynomial time, then $P=NP$
 - d. 'A' may be undecidable
3. A sorting technique is called stable if it
 - a. takes $O(1)$ times
 - b. maintains the relative order of occurrence of non-distinct elements
 - c. uses divide-and-conquer paradigm
 - d. takes $O(n)$ space
4. Correct choice of data structures can improve the performance of algorithms.
Match the following algorithms with appropriate data structures:

i. Breadth first search	a. Heap
ii. Depth first search	b. Stack
iii. Sorting	c. Queue

 - a. i-a ii-b iii-c
 - b. i-b ii-a iii-c
 - c. i-c ii-b iii-a
 - d. i-b ii-c iii-a
5. Which of the following is *not* a string matching Algorithm ?
 - a. Naïve String Algorithm
 - b. Rabin-Karp algorithm
 - c. Warshall's and Floyd's algorithm
 - d. Knuth-Morris-Pratt Algorithm

C Answer in One or Two Words.

1. What is Algorithm design strategy used in Quick Sort ?
2. What Algorithm design strategy is used in Prim's & Kruskal's Algorithms ?
3. What Algorithm design strategy is used in Graph coloring Problem ?
4. Principal of Optimality is related to which of Algorithm design strategy?
5. Write one example of NP Complete problem.

Q.2 Answer the following questions. (Attempt any three)

(15)

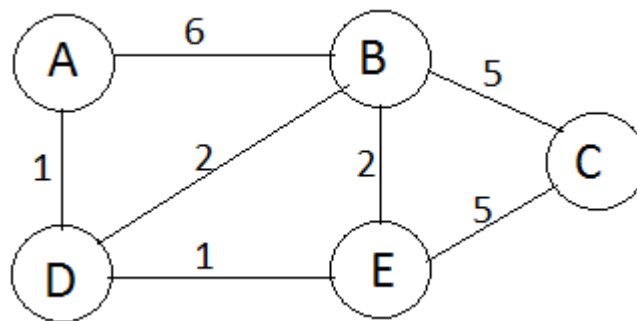
- A) Perform complexity analysis of Strassen's method of matrix multiplication.
- B) Differentiate between Backtracking and Branch & Bound strategies. Mention atleast 2 suitable examples of each.
- C) Given Pattern P = AAC & text T = ACAACA, find all positions where this pattern P occurs in text T using Naïve string matching algorithm.
- D) Differentiate between P & NP problems.

Q.3 A) Explain all three asymptotic notations with 2 examples of each.

(07)

- B) Given a graph and a source vertex "A" in graph, find shortest paths from source to all vertices in the below given graph using Dijkstra's shortest path algorithm.

(08)



OR

- B) Using greedy approach Obtain a set of optimal Huffman Codes for the following characters with their relative frequencies. Draw the decode tree for sets of codes with "0" for left arc and "1" for right arc.

(08)

Character	A	B	C	D	E	F
Frequency	2	3	5	7	9	13

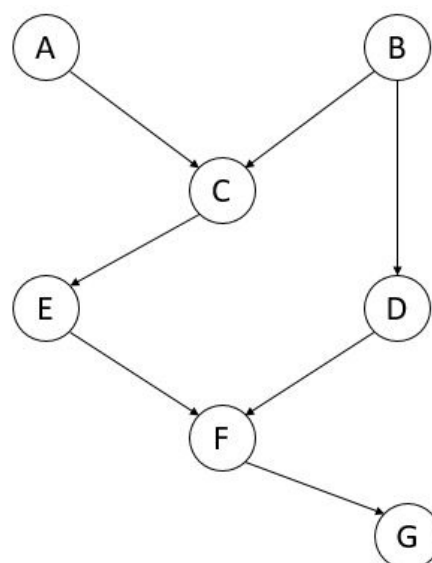
- Q.4 A) Write a short note on "n" queen's problem solution using backtracking method. Give solution for 8 queen problem.

(07)

OR

- A) Give Topological sorting order for following directed acyclic graph.

(07)



- B) Given Coins in different denominations of 1,5,7,8. How many minimum coins you would need to get value of 10 from this denomination. Use dynamic programming approach.

(08)